

"SOLANGE": Quantum Computing and AI Against Non-Druggable Mutations - A New Frontier in NSCLC Oncology

Scientific Oncology Legacy Advancing Undruggable Ground-state Energetics

BLAIS Challenge Award Proposal, August 2026

Project Aims

Lung cancer is the leading cause of cancer-related mortality worldwide. Approximately 60% of NSCLC patients harbor mutations with no approved targeted therapy - among them tumor-suppressor mutations in TP53, STK11, and KEAP1. These mutations are undruggable: their protein products lack binding pockets accessible to small-molecule drugs, leaving patients with few options and poor prognosis.

"SOLANGE" is a quantum-classical hybrid platform built to challenge this boundary. The project has three core aims:

1. **Scientific Aim:** Building on completed Phase 1 and Phase 2, advance computational modeling of undruggable NSCLC targets via quantum simulation - VQE (Variational Quantum Eigensolver) in Phase 3A, sqDRIFT sample-based quantum diagonalization on IBM Heron r3 in Phase 3B. Quantum necessity is established empirically, not assumed from system size: an automated classical-tractability classifier tests whether classical methods - including DMRG (density matrix renormalization group), run against the target's real, well-resolved crystal-structure site - already reach chemical accuracy before any target is prioritized for quantum simulation. In plain terms: quantum computing is used only where classical methods demonstrably fail.
2. **Translational Aim:** Bridge quantum computing and computational oncology, with interpretation and validation led by Professor Ray's group at KGI. Some of the most frequent (>50%) mutational lesions within TP53-encoded protein (p53) occur within its DNA-binding domain, a highly structured region with a well-resolved crystal structure. This project will evaluate the ground-state and interaction energetics needed to assess and prioritize candidate small-molecule stabilizers designed to restore or enhance mutant-protein function in tumor-associated mutant p53, a computational challenge currently outside classical computing abilities. Similar challenges arise for loss-of-function mutations in STK11 and KEAP1, where stabilizing a function-restoring agonist likewise requires quantum computing. This project assesses how each target's ground-state energetics inform stabilizer strategy. In plain terms: we search for a molecule to reactivate the broken tumor-suppressor gene.
3. **Collaborative Aim:** Professor Animesh Ray (W.M. Keck Foundation Professor of Systems, Computational, and Molecular Biology, KGI) contributes genomics and computational-interpretation expertise informing target prioritization, while CGU leads the quantum-classical platform, HPC orchestration, and governance architecture. In plain terms: CGU builds and runs the platform; KGI supplies the cancer-biology expertise to interpret it.

What Already Exists

"SOLANGE" arrives at submission as a functioning, deployed research platform, now validated on real quantum hardware. Existing assets include:

- **Live deployed platform** - operational at solange-platform.bio: processing real patient NGS lab reports (PDF/VCF) for tumor-suppressor mutations, running live VQE quantum simulations (Phase 3A), and displaying results across a six-tab research interface. Multi-user, cloud-deployed, access-controlled via individually credentialed Supabase Authentication.
- **Real quantum-hardware execution already achieved** - "SOLANGE" has dispatched and executed a live circuit on real IBM Heron r2 hardware and returned a verified, notarized result - not a simulator, not a dry run. This validates the Phase 3B execution pathway end-to-end, ahead of schedule.

- **Governed cross-boundary HPC orchestration** - a pull-based architecture routes jobs to USC's CARC "Laguna" cluster across a hard institutional security boundary without storing cluster credentials: an already-authenticated agent polls the queue outbound, executes, and returns results. Validated end-to-end with live HPC and DMRG runs.
- **Classical-tractability classifier** - an automated diagnostic determines, per target, whether classical computation (exact diagonalization or DMRG) reaches chemical accuracy before any quantum-necessity claim is made, measuring electronic entanglement against the mutation's real PDB crystal structure - an auditable, LEON-notarized classification.
- **Backend infrastructure** - a FastAPI (Python) REST API on Render, a React/TypeScript frontend on Netlify, and a Supabase PostgreSQL database handling authentication, simulation records, and provenance - all production-grade and continuously deployed.
- **3D quantum molecular visualizer** - real-time rendering of molecular targets and simulation states, built with Three.js.
- **A provenance & audit-trail layer** - designed to support 21 CFR Part 11: a nine-element (P1-P9) provenance record is generated for every run and independently re-verified by LEON, recomputing each record's cryptographic seal at ingestion and rejecting any mismatch. Every notarization and rejection is appended to a persistent, tamper-evident audit ledger.
- **Poster and architecture documentation** - a completed poster and technical block diagrams of the pipeline.

Methods and Approach

Computational Architecture

"SOLANGE" integrates four computational layers:

- **Quantum Layer (Phase 3A / 3B):** Phase 3A runs live VQE simulations using a 4-qubit Hamiltonian. Phase 3B executes on IBM Heron r3 hardware.
- **HPC Layer:** Classical reference calculations benchmark the quantum layer's outputs, and a classifier empirically determines, per target, whether classical methods reach chemical accuracy, establishing quantum necessity from measured evidence against the mutation's real crystal-structure active site.
- **AI Layer:** Machine learning models process mutation data, map patient mutations to target proteins, and rank simulation priorities, feeding into the compliance and provenance pipeline.
- **Governance Layer:** Every result crossing from an execution backend into "SOLANGE" is independently re-verified before it is trusted - verify-don't-trust, proven across classical HPC, real quantum hardware, and the classification module.

CGU–KGI Collaborative Research Model

CGU and KGI operate as an integrated transdisciplinary team: CGU leads the quantum-classical architecture, HPC execution, and governance framework; Professor Ray and KGI researchers contribute molecular-systems biology, genomics, and biological interpretation, evaluating NSCLC targets and results. BLAIS funding will create a joint CGU–KGI graduate research team in which students work across institutional and disciplinary boundaries rather than in parallel institutional silos. The teams meet monthly, jointly reviewing findings and co-authoring manuscripts and funding proposals. Students from both institutions work within SOLANGE's existing codebase - extending the LEON dashboard, running DMRG classifications on additional targets, and contributing to the mutation-data-to-simulation pipeline - each producing a reviewable artifact. For CGU, the project establishes a reusable quantum-AI research capability, a new KGI partnership, advanced doctoral opportunities,

joint scholarship, and positions the University for larger externally funded work in computational health and quantum-enabled science.

Schedule

Period	Milestones
Sep – Nov 2026	Partnership formalization; CGU–KGI student researcher onboarding; IBM hardware access secured; DMRG classifications completed across all core NSCLC targets
Dec 2026 – May 2027	Full-scale Phase 3B IBM Heron r3 sqDRIFT campaign at flagship target size; joint working groups
Jun – Sep 2027	Extended Phase 3B campaign across TP53 / STK11 / KEAP1; platform update integrating quantum-hardware results
Oct 2027 – Feb 2028	Final report; arXiv submission; CGU–KGI joint external funding proposal (NSF / NIH R01) developed and submitted

Expected Deliverable

The core deliverable is the **full-scale Phase 3B sqDRIFT campaign**: execution at flagship target size on IBM Heron r3, producing ground-state energy profiles for TP53, STK11, and KEAP1 undruggable mutations. IBM demonstrated comparable execution (32 electrons / 100 qubits, arXiv:2603.08696), placing targets within today’s demonstrated range - plausible, not guaranteed. BLAIS funding secures the IBM Heron r3 access this campaign requires.

Measures of Success:

- DMRG tractability classifications completed for at least 8 structurally-verified NSCLC point-mutation targets, 8 already completed and converged as of August 2026, establishing the classifier's real-world throughput.
- Phase 3B quantum executions completed for 2-3 targets meeting the classifier's own quantum-necessity criterion (measured $S_{\text{max}} \geq \text{threshold}$, not a size-based estimate)
- Quantum results benchmarked against classical DMRG calculations using the project's defined convergence criterion ($\Delta E < 1.6$ mHa between successive bond dimensions), ensuring every quantum execution is justified by real evidence of classical intractability.
- At least 2 CGU and KGI graduate students participating, each producing a reviewable research artifact.
- One joint CGU–KGI manuscript submitted, documenting the evidence-based classification framework and its application across the verified target set.
- SOLANGE updated to incorporate validated Phase 3B results and associated provenance records - every result LEON-sealed (P1–P9) and reproducible from the published pipeline, not a standalone claim.

Leadership

Itamar Shabtai (CGU) - Principal Investigator: leads project oversight, coordination of the CGU-KGI collaboration, integration of information systems, governance, AI, and translational-healthcare dimensions

Animesh Ray (KGI) - Co-Investigator leads biological interpretation of NSCLC targets and genomic analysis.

Molly Schmid (USC Marshall) - Scientific Advisor: former Princeton molecular biologist and biotech executive, NIH SBIR reviewer, advises on translational strategy and external funding pathways.

Doron Cohen (CGU) - PhD Candidate, SOLANGE Platform Lead: architects and builds the quantum-classical platform, governance layer, and HPC/quantum integration

Eligibility Alignment

BLAIS Mission - Intercollegiate Cooperation: "SOLANGE" establishes a new CGU-KGI research collaboration at the intersection of quantum computing, AI, computational biology, and oncology: CGU contributes quantum-classical computing, HPC, and governance expertise; KGI, through Professor Ray, contributes molecular biology, genomics, and biological interpretation. Faculty and students jointly evaluate targets, interpret results, and pursue follow-on funding - a capability neither institution could assemble alone.

CGU Strategic Research Priorities: The project targets Human Health & Flourishing - the most lethal, treatment-resistant cancer subtypes - and Data Analysis & Computational Mathematics, demonstrated today through SOLANGE's governance layer (LEON): cryptographic re-verification at ingestion, an append-only audit ledger, and orchestration crossing a hard institutional security boundary.

Applied and Transdisciplinary: "SOLANGE" processes real patient genomic data within an FDA-compliant framework, integrating quantum physics, oncology, and regulatory science. Developed under Design Science Research (DSR), "SOLANGE" is the evaluated artifact; the transferable contribution is its governance principles, validated through real-hardware operation.

External Funding Pathway: BLAIS provides the intercollegiate team structure and Phase 3B data that larger federal awards (NIH R01, NSF Quantum Leap Challenge Institutes) require as preliminary evidence - already supplied by "SOLANGE", validated on real hardware.